Package ‘scScale’

June 27, 2026

Title Single-Cell Scaling Law Utilities

Version 0.0.1

Author Jingyuan Hu [aut, cre]

Maintainer Jingyuan Hu <jingyuanhu@example.com>

Description Median Marchenko-Pastur noise calibration, BBP spike inversion, spectral overlap, and task-difficulty utilities for single-cell scaling analyses, aligned with ChesleaZ/scScale.

URL https://github.com/ChesleaZ/scScale

BugReports https://github.com/ChesleaZ/scScale/issues

Encoding UTF-8

Depends R (>= 2.10),
Matrix

LazyData true

LazyDataCompression xz

Suggests irlba,
SeuratObject

Roxygen list(markdown = TRUE)

RoxygenNote 7.3.2

Contents

gse123025_myeloid_hvg *GSE123025 Myeloid HVG Count Matrix Example*

Description

A larger example subset derived from GSE123025 for RMT spike-model tutorials. The object keeps all cells from the source matrix and restricts genes to 2,000 highly variable genes selected by `select_hvgs()`.

Usage

```
data(gse123025_myeloid_hvg)
```

Format

A list with:

counts Integer matrix with 2,000 selected genes/features by 1,922 cells.

source Source file name in the canonical lab data directory.

accession GEO accession identifier, `"GSE123025"`.

description Short description of the HVG subset.

selection List describing the feature and cell selection.

original_dim Dimensions of the source matrix before HVG selection.

cell_totals Column sums of the HVG count matrix.

gene_totals Row sums of the HVG count matrix.

Source

Derived from `data/GSE123025/GSE123025_Single_myeloid_1922_cells_processed_data.csv.gz` in the canonical lab workspace. The source matrix has 23,429 genes by 1,922 cells and is about 8.0 MB compressed as CSV gzip, or 95 MB uncompressed.

Examples

```
data(gse123025_myeloid_hvg)
counts <- gse123025_myeloid_hvg$counts
dim(counts)
fit <- scscale_fit(counts, r = 10, mp_max_iter = 3, fit_umi = FALSE)
fit
```

gse164378_3p_citeseq_hvg
GSE164378 3 Prime CITE-seq RNA HVG and ADT Example

Description

A compact paired RNA and ADT example derived from the 3 prime CITE-seq portion of GSE164378. The object keeps 4,000 matched PBMC cells, 2,000 RNA highly variable genes, and all 228 measured ADT features.

Usage

```
data(gse164378_3p_citeseq_hvg)
```

Format

A list with:

rna_counts Sparse integer matrix with 2,000 selected RNA genes by 4,000 cells.

adt_counts Sparse integer matrix with 228 ADT features by the same 4,000 cells.

meta Cell metadata, including RNA/ADT QC totals, lane, donor, time, batch, and cell type labels.

source Short source dataset description.

accession GEO accession identifier, `"GSE164378"`.

description Short description of the compact paired-modality subset.

selection List describing the cell and RNA feature selection.

original_dim Dimensions of the source 3 prime RNA and ADT matrices before subsetting.

cell_totals RNA and ADT column sums for the selected cells.

feature_totals RNA and ADT row sums for the selected features.

Source

Derived from the 3 prime RNA and ADT matrices in GSE164378, the Hao et al. multimodal PBMC reference dataset. The source 3 prime matrices have 33,538 RNA features by 161,764 cells and 228 ADT features by the same 161,764 cells.

Examples

```
data(gse164378_3p_citeseq_hvg)
rna <- gse164378_3p_citeseq_hvg$rna_counts
adt <- gse164378_3p_citeseq_hvg$adt_counts
dim(rna)
dim(adt)
```

scScale *Single-Cell Scaling Law Utilities*

Description

Object-style Gaussian spike fitting and spectral mutual information utilities for single-cell scaling-law analyses.

Usage

```
scscale_fit(x, input = c("counts", "normalized", "representation",
  "eigenvalues"), target = NULL, target_fit = NULL, n = NULL, p = NULL,
  target_depth = 1e4, center = TRUE, scale = TRUE, n_features = NULL,
  min_cells = 10, mp_max_iter = 50, mp_grid_n = 20000,
  mp_stop_metric = c("qq_rmse_log", "ks", "neg_log_likelihood", "none"),
  mp_stop_tol = 1e-4, mp_min_iter = 2, r = NULL,
  fit_umi = input[1] == "counts",
  sampling_rates = c(0.10, 0.20, 0.35, 0.50, 0.70, 0.85, 1.00),
  U_grid = NULL, umi_seed = 1, umi_replicates = 1,
  store_matrix = FALSE, use_irlba = TRUE)

scscale_mi(fit, target_fit, n_grid = NULL, U_grid = NULL,
  sampling_rates = NULL, combine = TRUE, empirical = TRUE,
  store_empirical_subspaces = FALSE, use_irlba = TRUE, r = NULL,
  eps = 1e-12)
```

Arguments

`x` A feature-by-cell count matrix, sparse **Matrix**, normalized feature-by-cell matrix, low-dimensional representation, or covariance eigenvalue vector.

`input` Input type. `"counts"` applies library-size normalization, `log1p`, and row standardization. `"eigenvalues"` requires `n` and `p`.

`target` Optional empirical target.

`target_fit` Optional target-side `scscale_fit` object.

`n, p` Cell and feature counts, required when fitting directly from eigenvalues.

`target_depth`, `center`, `scale`, `n_features`, `min_cells`
Preprocessing settings for count matrices.

`mp_max_iter`, `mp_grid_n`, `mp_stop_metric`, `mp_stop_tol`, `mp_min_iter`
MP bulk fitting controls.

`r` Number of spikes or singular vectors to keep.

`fit_umi` Whether `scscale_fit()` should also estimate UMI scaling curves.

`sampling_rates`, `U_grid`
Sequencing-depth grid as global sampling rates or expected total UMI counts.

`umi_seed`, `umi_replicates`
Random seed and replicate count for stochastic UMI downsampling.

`store_matrix` Whether to store the normalized matrix inside the returned fit object.

`use_irlba` Whether to use **irlba** for partial singular-vector computations when available.

`fit`	An `scscale_fit` object for the measured modality.
`n_grid`	Optional cell-number grid for MI scaling.
`combine`	If `TRUE`, return the full cell-number by UMI grid when both grids are supplied to `scscale_mi()`.
`empirical`	Whether to compute empirical subspace MI from stored normalized matrices. Both fitted objects must have been created with `store_matrix = TRUE`.
`store_empirical_subspaces`	Whether to keep the fitted RNA and target right singular vector matrices in the `empirical` output component.
`eps`	Numerical clipping value for MI calculations.

Details

The public workflow is:

1. Fit each modality with `scscale_fit()`.
2. Use one fitted object as the target in `scscale_mi()`.
3. Optionally provide cell-number and/or UMI grids to evaluate scaling laws.

Value

`scscale_fit()` returns an `scscale_fit` object with dimensions, normalization metadata, covariance spectrum, MP bulk fit, spike table, finite and infinite recoverability vectors, optional UMI scaling, and optional MI results. `scscale_mi()` returns theoretical MI, infinite-cell MI, empirical subspace MI when stored matrices are available, major intermediate quantities such as `q_X`, `d2_X`, and theta products, and, when requested, UMI, cell-number, or full cell-by-UMI scaling tables.

Examples

```
data(gse164378_3p_citeseq_hvg)
rna <- gse164378_3p_citeseq_hvg$rna_counts[, 1:200]
adt <- gse164378_3p_citeseq_hvg$adt_counts[, 1:200]

fit_rna <- scscale_fit(rna, r = 3, fit_umi = FALSE, store_matrix = TRUE,
  mp_grid_n = 500)
fit_adt <- scscale_fit(adt, r = 3, fit_umi = FALSE, store_matrix = TRUE,
  mp_grid_n = 500)
mi <- scscale_mi(fit_rna, fit_adt, n_grid = c(100, 200), r = 3)
mi$I_empirical
mi$cell_scaling
```

Index